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ASSOCIATION OF REPETITIVE ELEMENTS AT CHEMOSENSORY RECEPTOR GENES IN *HYDROPHIS* GENOMES

CONTEXT. *Hydrophis* sea snakes diverged from terrestrial Australian elapids 9-18 MYA, and have since rapidly radiated, almost simultaneously 1 MYA, into the current 48 species (Krishida et al. 2025; Ludington et al. 2023; Sanders et al. 2008). With this compressed radiation came significant selective pressures imposed by the marine environment. Notably, the change in chemosensory medium, from air to water. Unlike mammals, reptiles (Squamata) retain few vomeronasal type-1 receptor genes but have large, lineage-expanded vomeronasal type-2 receptor (V2R) repertoires; V2Rs detect water-soluble cues delivered to the vomeronasal organ by tongue-flicking (Brykczynska et al. 2013). Chemosensory receptor copy number variation arises through repeated gene gain and loss and can contribute to ecological adaptation (Niimura and Nei 2006; Hughes et al. 2018). Preliminary analysis found substantial repeat content between V2R loci arranged in tandem arrays, motivating a formal test of whether this architecture exceeds genomic background.

BACKGROUND

Long Terminal Repeat Retrotransposons

Long terminal repeat retrotransposons (LTRs) are a mechanistically plausible contributor to copy-number change. Similar repeats in direct orientation can act as substrates for non-allelic homologous recombination (NAHR), duplicating or deleting the intervening sequence (Hastings et al. 2009), while recombination within LTR retrotransposons can leave solo LTRs and truncated fragments (Devos et al. 2002; Ma et al. 2004). Breakpoint evidence exists in mammalian KIR immune-gene clusters, where copy-number and hybrid-gene formation map to an MLT1 LTR fragment (Traherne et al. 2010). Comparative studies provide relevant but not causal precedents: transposable elements mediated V2R-containing gene traffic between chromosomes in *Varanus acanthurus* (a lizard; Zhu et al. 2026); a *Vipera ammodytes* viper genome contains abundant repeats and expanded chemosensory repertoires (Rao et al. 2025); and younger transposable elements are enriched near chemosensory receptors in aphids (Olvera-Vazquez et al. 2025). Transposable elements can also donate promoters, enhancers, transcription-factor binding sites and regulatory RNAs (Chuong et al. 2017). Preliminary annotation of reassembled *H. ornatus* reveals concentrated LTR content at V2R cluster intervals, motivating formal spatial testing.

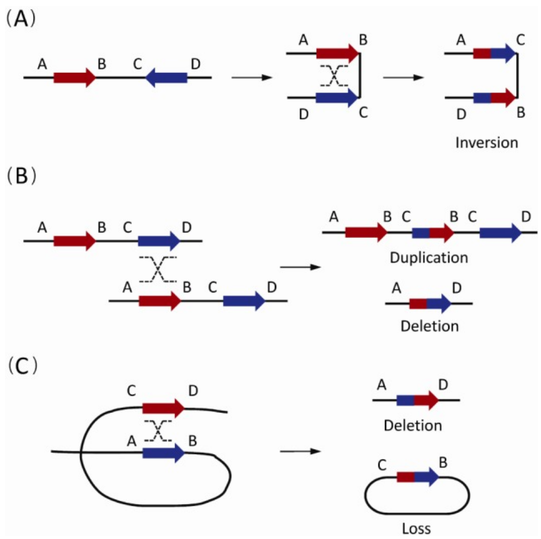

Figure 1. General outcomes of repeat-mediated non-allelic homologous recombination (NAHR). Reverse-oriented repeats cause inversions (A), while directly oriented repeats produce inter-chromatid duplications or deletions (B) and intra-chromatid deletions with loss of circular DNA (C). Reproduced from Chen et al. (2014)

Genome Complexity

Squamate genomes show substantial lineage-specific variation in repetitive-element content (Pasquesi et al. 2018). Reptile sex chromosomes can accumulate repeats under recombination suppression:

the rattlesnake W is a retroelement-rich chromosome (Schield et al. 2022), while Varanid lizard ZW microchromosomes differentiated through rapid repeat amplification (Matsubara et al. 2014). Repeat-rich tandem arrays are also difficult to assemble, therefore, mappability and the LTR Assembly Index (LAI) provide complementary measures that represent how well sequence has been resolved in complex regions (Pockrandt et al. 2020; Ou et al. 2018). In *Hydrophis*, CZ carries the highest structural-variant burden of any chromosome (Ludington et al. 2023), and C2 and CZ contain most V2Rs in the present annotation. These chromosomes were therefore analysed separately rather than pooling across distinct evolutionary and technical contexts.

Existing reptile studies describing repeat-rich chromosomes and chemosensory repertoires do not test V2R–LTR association against a spatial null (Zhu et al. 2026; Rao et al. 2025). Descriptive co-localisation cannot separate local enrichment from background genome structure because genomic features are non-uniformly distributed (Heger et al. 2013).

RESEARCH AIMS

Two hypotheses are tested for the association between LTRs and chemosensory (V2R) expansion. **H1 (spatial association)**: LTR coverage at V2R loci and clusters exceeds a chromosome spatial null, controlled for chromosomal context (e.g., GC-content matching). **H2 (dose-response):** Expected V2R cluster paralog count increases with local LTR density.

These questions are addressed using permutation-based overlap testing, which builds a null distribution by shuffling gene positions thousands of times to define "chance" expectations. The core technical challenge is spatial pseudoreplication: because V2Rs are arranged in tandem clusters, individual loci are not independent data points. To solve this, cluster-level randomisation and cyclic permutation are implemented, which preserve the local architecture of gene arrays while testing their broader association with the repeat landscape.

However, as this report uses a single-genome inference without temporal ordering of LTR insertion times and V2R duplications, it can estimate spatial association and conditional dose-response but cannot establish causation.

DATA GENERATION

One reassembled *Hydrophis ornatus* genome was used to offer a higher-quality reference compared to published genomes for the species. Statistical tests were restricted to C2 and CZ. Repeat annotation was performed with `EDTA` for structural identification and `RepeatMasker` for homology-based classification, producing 261,347 LTR intervals on C2 and 148,232 on CZ (Ou et al. 2019; Tarailo-Graovac and Chen 2009). Three nested V2R confidence sets were defined. Candidate loci were located via `tBLASTn` targeting the conserved seven-transmembrane domain and independently validated against a profile Hidden Markov Model (`pHMM`) built from curated vertebrate V2R sequences using `HMMER` (Altschul et al. 1997; Eddy 2011). The primary conservative set requires structural intactness confirmed by `pHMM` and independent `tBLASTn` confirmation: 157 loci (112 on C2; 45 on C2). A broader candidate set (1,681 loci) serves as a sensitivity comparator.

ANALYTICAL APPROACH & METHODS

Null Hierarchy

To ensure rigour in experimental design and its conclusions, a nested hierarchy of null models were used:

- **Null A**: shuffles V2R intervals genome-wide. This test is biologically invalid as V2Rs are non-randomly positioned on specific chromosomes.
- **Null B**: shuffles intervals only within CZ and C2 (independently), controlling for distinct structural repeat and regulatory landscapes.
- **Null C**: GC-matched resampling from candidate positions binned by GC-content. This controls for GC-bias of both LTRs and chemosensory receptor arrays.
- **Null D**: further restricts the workspace by excluding low-mappability and coverage-anomaly regions (ref).
- **Cyclic permutation**: rather than shuffling V2R positions, the chromosome's LTR track is rotated by a random circular offset while V2R loci remain fixed (Rapoport et al. 2024). This preserves the spatial autocorrelation (natural repeat clumping) of the LTR landscape: the most conservative null.

Universe

The set of positions within which permuted V2R intervals may land. Defined as the chromosome complement of assembly gaps (N-runs), low-mappability regions (low Shannon entropy, measured with `GenMap`; Pockrandt et al. 2020) and coverage-anomaly 1 Mb bins. Restricting the universe prevents permuted loci from occupying regions where no real gene could be reliably called, which would lower the null mean and artificially inflate enrichment. The universe is constructed independently for C2 and CZ.

$$M_i = \frac{1}{C(s,k,e)} \tag{1}$$

Mappability: s is the sequence; k is the k-mer length; e is the allowed number of mismatches; and $C(s,k,e)$ counts occurrences of that k-mer. The equation measures sequence uniqueness: M_i is 1 for a unique k-mer and decreases as the number of matching copies increases.

Overlap

The measure of LTR–V2R co-occurrence. Four measures capture different aspects of LTR–V2R co-occurrence: `numOverlaps` (presence/absence; Gel et al. 2016), `meanDistance` (proximity; Gel et al. 2016), Jaccard index (base-pair intersection divided by union), and flank-ring coverage (flanking LTR sequence excluding the gene body). Each statistic is interpreted against its own permutation distribution because their fold-enrichments have different denominators and are not directly comparable.

$$Jaccard = \frac{|\mathrm{LTR} \cap \mathrm{V2R}|}{|\mathrm{LTR} \cup \mathrm{V2R}|} \tag{2}$$

Jaccard Index: LTR and V2R are the sets of base pairs covered by each feature. The equation measures proportional overlap from 0 (none) to 1 (identical coverage).

Flank

The distance around each V2R within which LTRs are counted. The spatial profile (Figure 2) shows elevated repeat density within 25 kb and a marked decline by 50 kb, motivating a sweep across 1, 5, 10, 25 and 50 kb flank sizes.

This decay is consistent with a local association, while flanking homologous repeats provide the architecture required for NAHR-mediated copy-number change (Hastings et al. 2009; Traherne et al. 2010).

PERMUTATION ENRICHMENT TEST

Enrichment of LTRs at V2Rs was tested using the `regioneR` package, its `permTest`, `randomizeRegions` (Gel et al. 2016; $n = 10,000$ permutations) and custom evaluation functions, which shuffle V2R positions, while preserving their number and length distribution.

The null hierarchy underscores the cost of ignoring chromosomal context. **Null A** (genome wide) produces 3.64-fold enrichment ($z = 65.16, p < 0.0001$) for all repeats against the broad V2R set. This is a biologically invalid test, as described above, due to non-random distribution of V2Rs in clusters.

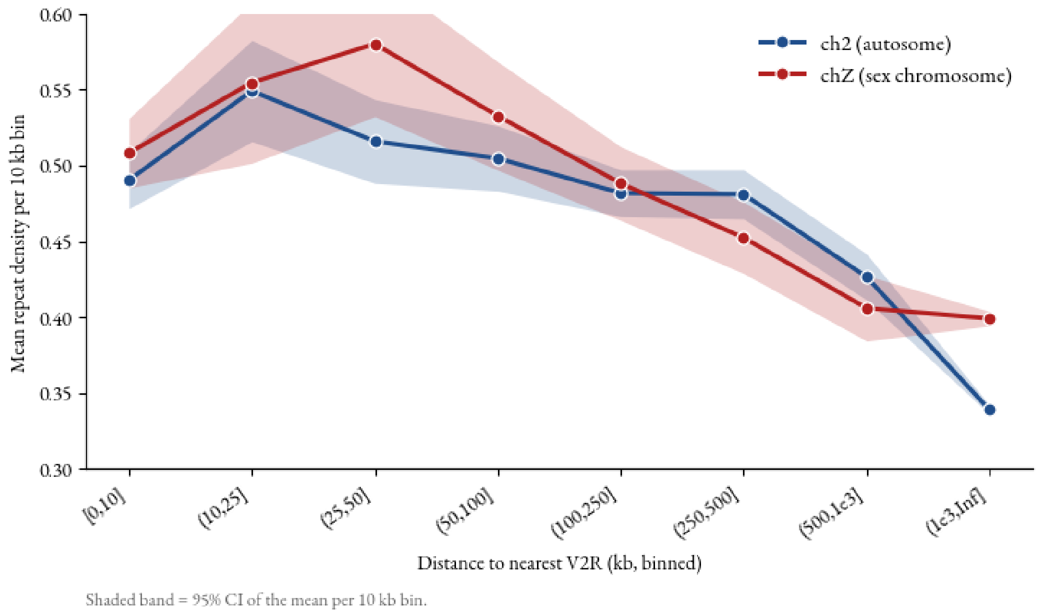

Figure 2: Proximal enrichment and distance decay. Mean repeat density per 10kb bin (±95% CI) against binned distance to the nearest V2R, per chromosome.

Null B restricts shuffling of the primary V2R set to C2/CZ, collapsing this fold to 1.55 (C2) and 1.40 (CZ) ($p < 0.05$). This significant reduction in enrichment supports the premise that genome-wide permutations inflate signal when features have structured chromosomal placement (Heger et al. 2013).

CLUSTER-LEVEL & PSEUDOREPLICATION CORRECTION

Spatial pseudoreplication is the dominant threat to the experiment's validity. In per-locus permutation tests of tandem gene families, paralogs separated by 2 kb re-sample the same chromosomal neighbourhood, inflating effective sample size and z-statistics (Waples et al. 2022). Adjacent loci were therefore merged into clusters at a 100 kb primary merge distance (motivated by the nearest-neighbour gap distribution in Figure 3), and permutations were performed by randomising whole clusters as units.

The 100 kb merge threshold for cluster size was chosen from the nearest-neighbour gap distribution of the V2R set. Inter-paralog gaps within tandem arrays range from 2.7 to 97.6 kb (median 33.4 kb, $n = 33$ gaps); 100 kb captures all within-array pairs while leaving inter-array distances (median 453 kb) unmerged. Merge distances of 25, 50 and 100 kb were qualitatively identical across all three, confirming that results are not artefacts of this selection.

LTR density was calculated as the proportion of each V2R cluster covered by LTRs, giving a continuous measure of extent, not binary presence. Clusters are biologically coherent units because clustered olfactory genes can share transcriptional regulation (Dietschi et al. 2022). Unlike Jaccard, which penalises LTR sequences outside the gene, this is not relevant once the cluster window is already defined.

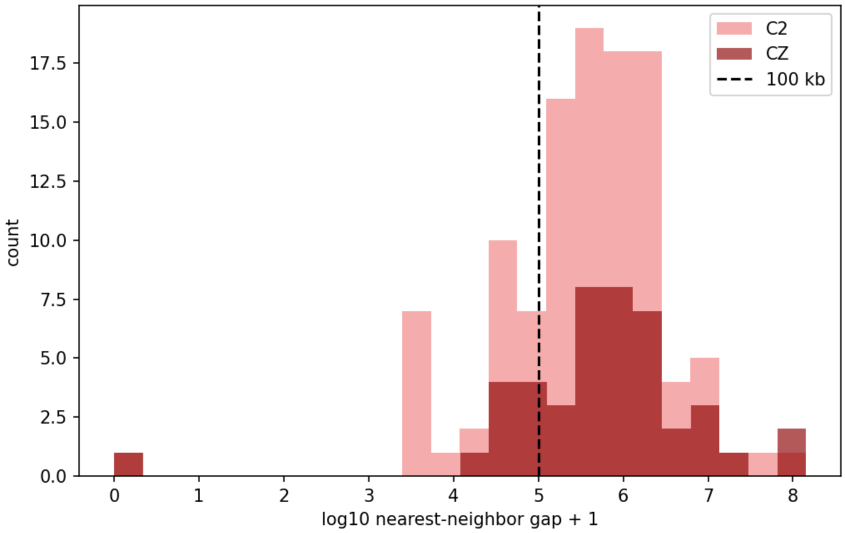

Figure 3: Nearest-neighbour gaps between consecutive V2Rs (n = 155 pairs). The distribution is bimodal: a smaller left mode separates paralogs inside the same tandem array (2.7–97.6 kb, median 33.4 kb, $n = 33$), while a larger mode separates independent V2R arrays (overall median gap 453 kb). The 100 kb merge threshold sits between these modes, joining all within-array paralogs while leaving distinct arrays unmerged.

Coverage Fraction

Cluster-level enrichment was evaluated across four nulls. Under chromosome-matched **Null B**, C2 showed fold = 1.39 ($z = 6.77$, $p < 0.001$) and CZ fold = 1.20 ($z = 2.18$, $p = 0.017$). **Null C** additionally draws permuted clusters from the same chromosome-specific GC bin, following the match-then-test logic of `matchRanges` (Davis et al. 2023). Significance remains on C2 (fold = 1.37, $z = 7.04$, $p < 0.001$) and CZ (fold = 1.20, $z = 2.27$, $p = 0.013$). **Null D** excludes low-mappability and coverage-anomaly regions; enrichment remains on C2 (fold = 1.59, $z = 11.26$) and CZ (fold = 1.48, $z = 9.30$; both $p < 0.001$).

$$\text{Coverage Fraction} = \frac{|\text{LTR} \cap W|}{|W|} \tag{3}$$

Coverage fraction: W is the merged V2R-cluster window; $\text{LTR} \cap W$ is the part of that window covered by LTRs. The equation calculates the proportion of each cluster occupied by LTR sequence, from 0 (no coverage) to 1 (complete coverage).

Cyclic Permutation

Cyclic permutation was implemented using `GenomicRanges` to rotate the chromosome's repeat track, following the approach of Rapoport et al. (2024) rather than shuffling V2R positions, thus preserving the natural LTR landscape (autocorrelation) (Lawrence et al. 2013; Rapoport et al. 2024). C2 remains significant (fold = 1.36, $z = 3.21$, $p = 0.006$, $n = 2,000$ rotations). CZ is not statistically significant (fold = 1.24, $p = 0.090$), although enrichment still appears present. CZ signal was more method-sensitive, likely due to greater assembly complexity and poor quality at the cluster.

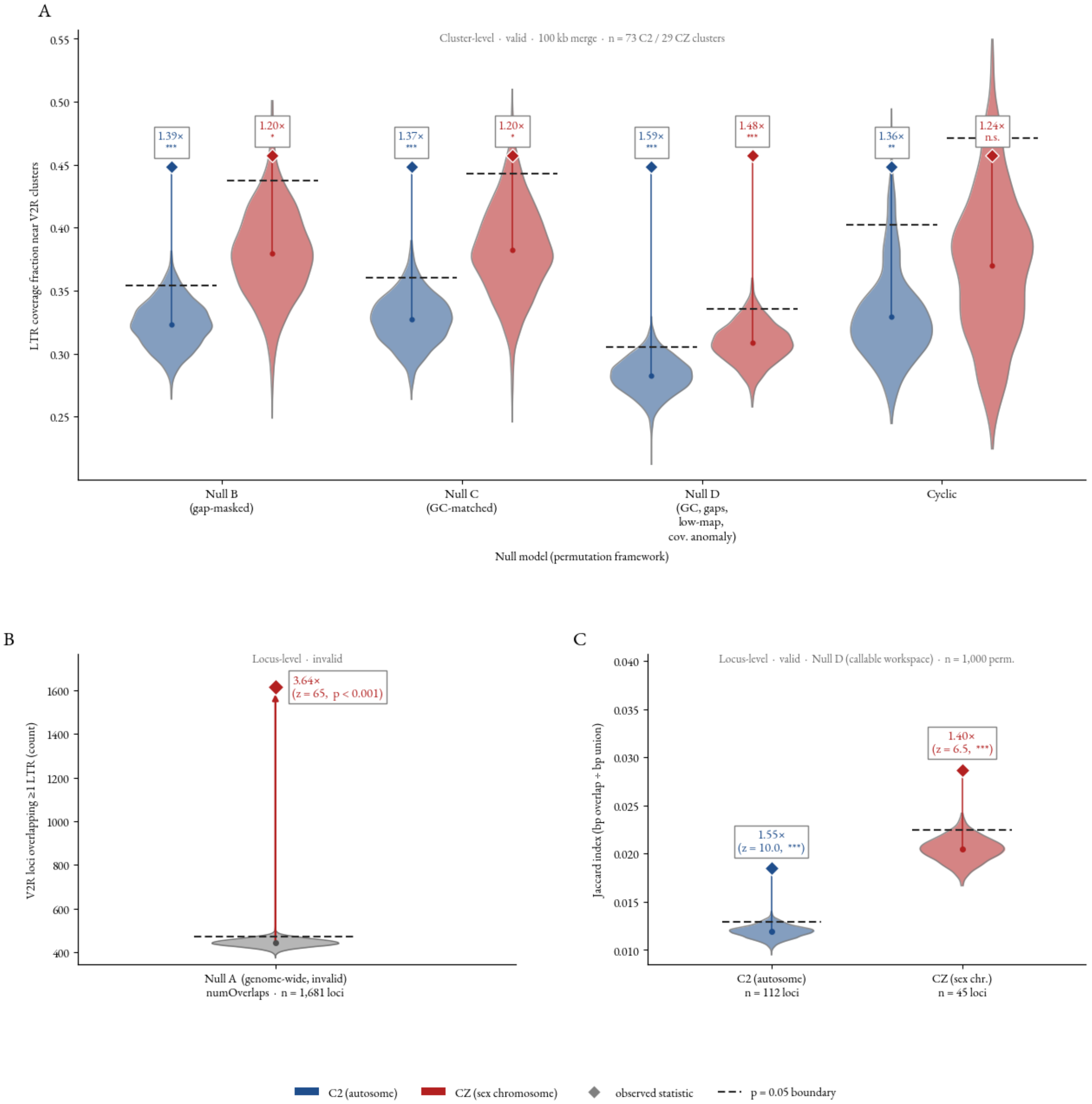

Figure 4: A: valid cluster-level permutation hierarchy (LTR bp coverage fraction, 100 kb merge). B: invalid locus-level Null A foil (genome-wide; `numOverlaps` count; shows chromosome-confound artefact). C: valid locus-level Null D (Jaccard bp index; primary V2R set, per chromosome). Note the null distribution in Panel C is approximated as $N(\text{null\_mean}, \text{null\_sd})$, back-calculated from the reported z-score ($n = 10{,}000$ permutations).

DOSE-RESPONSE MODELLING

To address **H2**, a Negative Binomial Generalised Linear Model (NB-GLM) was employed to determine if the number of V2R paralogs within a cluster scales with local LTR density. An NB-GLM was preferred over a baseline Poisson model because chemosensory gene counts typically exhibit overdispersion (where variance exceeds the mean). To statistically justify this transition, a Cameron–Trivedi test was performed (Cameron and Trivedi 1990).

$$\frac{(y_i - \hat{\mu}_i)^2 - y_i}{\hat{\mu}_i} = \alpha\hat{\mu}_i + u_i \tag{4}$$

Cameron-Trivedi test. $\hat{\mu}_i$ Poisson predicted mean; α: dispersion coefficient; u_i: unexplained error. Tests whether count variance exceeds Poisson expectation.

Unlike standard Likelihood Ratio Tests, which face parameter-boundary limitations (*e.g., when overdispersion is zero*), the Cameron–Trivedi test evaluates variance by regressing squared Poisson residuals on fitted values. A significant, positive coefficient from this test confirmed overdispersion, validating the NB-GLM.

$$n_{\text{loci}} \sim \text{LTR Density} + \text{Chromosome} + \text{GC-content} \tag{5}$$

Across reasonable combinations of V2R set, LTR class, flank and window, no robust monotonic relationship was detected. Local repeat density does not reliably predict larger paralog arrays. One specification reached nominal significance (C2, 5kb flank size: $p = 0.037$, $q = 0.59$), but none survived multiple-testing correction via the Benjamini–Hochberg (BH) procedure (smallest $q = 0.18$; Benjamini and Hochberg 1995), applied by ranking p-values to define a flexible threshold (q). This procedure controls the false discovery rate with greater power than family-wise Bonferroni correction, although dependence among overlapping genomic windows means adjusted values require caution (Benjamini and Yekutieli 2001).

Broad and conservative V2R sets were analysed separately because they differ in annotation reliability and count patterns. Results were compared across flank and cluster definitions to test robustness, focusing on effect direction, uncertainty and consistency rather than a single p-value.

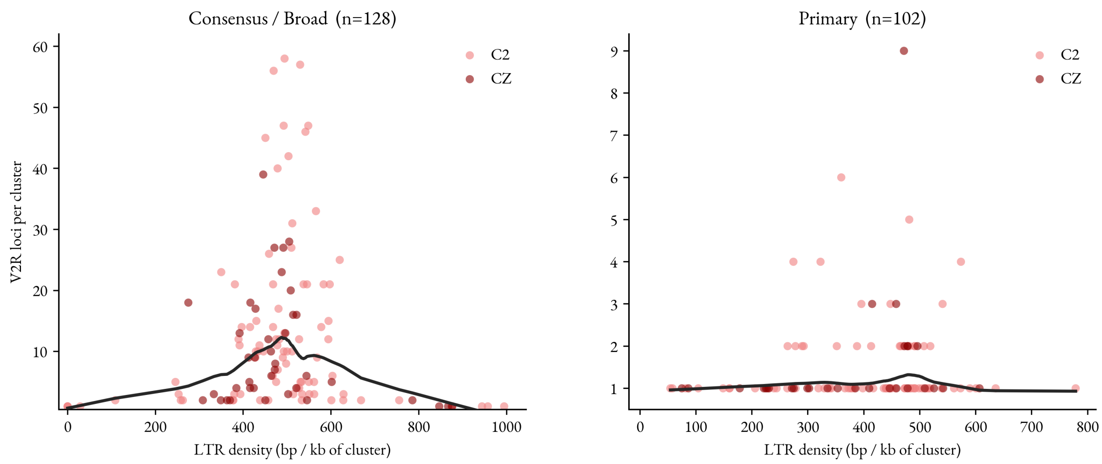

Figure 5: V2R paralog count against LTR density for the broad candidate set (left) and primary conservative set (right), stratified by chromosome. Fitted trends are weak and do not support a robust monotonic dose-response.

ALTERNATIVE METHODS & LIMITATIONS

Permutation Testing

Numerous alternative permutation testing tools were considered. `GAT` supports isochore and GC-stratified interval testing (Heger et al. 2013), but `regioneR` was preferred for its custom evaluation functions and local Z-score diagnostics (Gel et al. 2016). `bedtools shuffle` preserves interval lengths efficiently but lacks an integrated inference framework (Quinlan and Hall 2010). The `nullranges` alternatives provide blockbootstrap nulls `bootRanges` and multivariable covariate matching (`matchRanges`), offering stronger control of autocorrelation and residual GC imbalance than the present method's bins: a noted limitation (Mu et al. 2023; Davis et al. 2023).

These alternatives address different failure modes and are not interchangeable. Covariate matching improves balance but does not remove spatial dependence; block bootstrap preserves local dependence but may not balance GC content; cyclic rotation preserves the observed repeat landscape but introduces chromosome-end wrapping. A future benchmark should apply the same cluster statistic under each null and use simulation to compare the false-positive rate and power before selecting a primary design.

Annotation Constraints

V2R loci were defined using homology proxies rather than experimental validation. Local non-V2R gene density was excluded as a covariate, leaving a recognised confounding gap. Furthermore, standard tools like `EDTA` may fail to assemble complete LTR sequences in complex arrays, evident in `RepeatMasker's` 'texttttLTR/unknown' class, comprising 33.7% of the total annotated (likely fragmented LTRs: Figure 6), yet accounting for essentially all V2R enrichment (fold = 2.93, versus Gypsy = 0.12, Copia = 0.77). This limits our ability to distinguish true biological fragmentation (*i.e., solo-LTR fossils generated by purifying selection*) from technical annotation failures. `TEtrimmer` could improve consensus library curation, while `TEsorter` classifies conserved domains; both provide stronger validation than accepting `LTR/unknown` as a biological superfamily (Qian et al. 2025; Zhang et al. 2022).

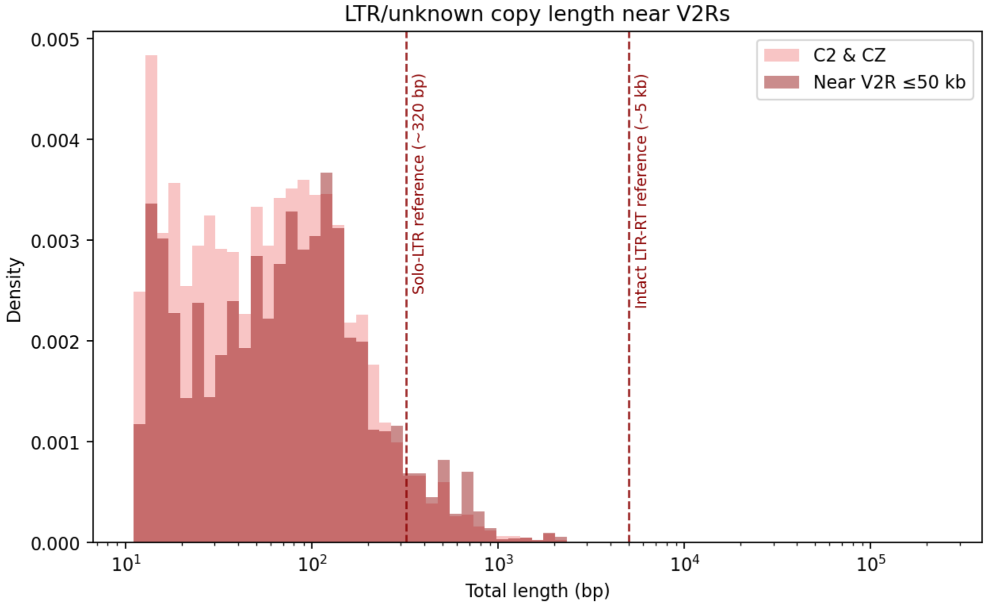

Figure 6: Length distribution of`LTR/UnknownRepeatMasker` copies genome-wide on C2/CZ versus within 50 kb of V2Rs. Dashed lines mark a solo-LTR-scale reference (*i.e., the terminal repeat size left after internal sequence excision: 320 bp*), and an intact LTR retrotransposon-scale reference (5 kb); most LTRs annotated in the class are likely fragmented (Devos et al. 2002; Ma et al. 2004).

Assembly Quality

The most severe limitation is the underlying assembly quality, evident in 88.2% of V2R-containing bins residing in low-mappability sequence. At 1 Mb resolution, V2R-containing bins exhibit depressed mappability (0.219 versus 0.536 background), elevated read depth deviation (0.567 versus 0.059), and a depressed LAI (16.83 versus 25.06). A model incorporating distance, GC content, and mappability could determine whether the repeat-density gradient is entangled with assembly quality. This limitation severely affects the CZ, where sequence collapse due to assembly and biological signal cannot yet be separated.

Inferential & Statistical Limitations

Colocalisation establishes spatial correlation but cannot prove causality. LTRs may actively drive duplication or suppression of recombination, or they might passively accumulate in regions of reduced recombination (Chen et al. 2023; Duhamel et al. 2023). Our reliance on single-genome inference prevents cross-species validation. The statistical framework also lacks formal power simulations and false positive rate calibration, which are essential for benchmarking spatial tests. Furthermore, while cluster-level aggregation reduced pseudoreplication, cluster-to-cluster spatial dependence remains unmodeled in the enrichment permutations.

Separating genuine evolutionary mechanisms from technical artifacts requires a robust biological null model. Future frameworks must incorporate simulated power analyses, advanced repeat annotation tools, and multi-genome datasets to overcome these methodological limitations.

CONCLUSION

Investigating V2R-array expansion in sea snakes requires separating the local evolutionary signal from chromosome-structure and assembly artefacts. Genome-wide shuffling substantially inflated enrichment relative to chromosome-matched nulls. Cyclic permutation supported localised LTR enrichment at V2R clusters on C2, whereas evidence on CZ was method-sensitive and strongly affected by complexity. No robust monotonic relationship between LTR density and array size was detected. The enriched short LTR annotations may be degraded structural remnants or annotation artefacts rather than active agents of duplication, underscoring the necessity of matched spatial null modelling in comparative genomics.

DATA & METHOD AVAILABILITY:

Scripts for all methods, plotting and data processing are available at https://github.com/blytrm/Hydrophiinae-genomics/tree/main/3_repeats/LTR-density_atV2Rs

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Word Count: 2850

REFERENCES

[1] Altschul SF, Madden TL, Schäffer AA, Zhang J, Zhang Z, Miller W, Lipman DJ (1997) Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. *Nucleic Acids Research* 25(17), 3389–3402.

[2] Benjamini Y, Hochberg Y (1995) Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society Series B* 57(1), 289–300.

[3] Benjamini Y, Yekutieli D (2001) The control of the false discovery rate in multiple testing under dependency. *The Annals of Statistics* 29(4), 1165–1188.

[4] Brykczynska U, Tzika AC, Rodriguez I, Milinkovitch MC (2013) Contrasted evolution of the vomeronasal receptor repertoires in mammals and squamate reptiles. *Genome Biology and Evolution* 5(2), 389–401.

[5] Cameron AC, Trivedi PK (1990) Regression-based tests for overdispersion in the Poisson model. *Journal of Econometrics* 46(3), 347–364.

[6] Chen M, Huang X, Wang C, Wang S, Jia L, Li L (2024) Endogenous retroviral solo-LTRs in human genome. *Frontiers in Genetics* 15, 1358078.

[7] Chuong EB, Elde NC, Feschotte C (2017) Regulatory activities of transposable elements: from conflicts to benefits. *Nature Reviews Genetics* 18(2), 71–86.

[8] Davis ES, Mu W, Lee S, Dozmorov MG, Love MI, Phanstiel DH (2023) matchRanges: generating null hypothesis genomic ranges via covariate-matched sampling. *Bioinformatics* 39(5), btad197.

[9] Devos KM, Brown JKM, Bennetzen JL (2002) Genome size reduction through illegitimate recombination counteracts genome expansion in *Arabidopsis*. *Genome Research* 12(7), 1075–1079.

[10] Dietschi Q, Tuberosa J, Fodoulian L, Boillat M, Kan C, Codourey J, Pauli V, Feinstein P, Carleton A, Rodriguez I (2022) Clustering of vomeronasal receptor genes is required for transcriptional stability but not for choice. *Science Advances* 8(46), eabn7450.

[11] Duhamel M, Hood ME, Rodríguez de la Vega RC, Giraud T (2023) Dynamics of transposable element accumulation in the non-recombining regions of mating-type chromosomes in anther-smut fungi. *Nature Communications* 14, 5692.

[12] Eddy SR (2011) Accelerated profile HMM searches. *PLoS Computational Biology* 7(10), e1002195.

[13] Gel B, Díez-Villanueva A, Serra E, Buschbeck M, Peinado MA, Malinverni R (2016) regioneR: an R/Bioconductor package for the association analysis of genomic regions based on permutation tests. *Bioinformatics* 32(2), 289–291.

[14] Hastings PJ, Lupski JR, Rosenberg SM, Ira G (2009) Mechanisms of change in gene copy number. *Nature Reviews Genetics* 10(8), 551–564.

[15] Heger A, Webber C, Goodson M, Ponting CP, Lunter G (2013) GAT: a simulation framework for testing the association of genomic intervals. *Bioinformatics* 29(16), 2046–2048.

[16] Hughes GM, Boston ESM, Finarelli JA, Murphy WJ, Higgins DG, Teeling EC (2018) The birth and death of olfactory receptor gene families in mammalian niche adaptation. *Molecular Biology and Evolution* 35(6), 1390–1406.

[17] Kishida T, Keboushi R, Sasai T, Toda M (2025) Genomics reveals recent rapid speciation of sea snakes of the genus *Hydrophis* (Reptilia, Squamata, Elapidae). *Ecology and Evolution* 15(6), e71627.

[18] Lawrence M, Huber W, Pagès H, Aboyoun P, Carlson M, Gentleman R, Morgan MT, Carey VJ (2013) Software for computing and annotating genomic ranges. *PLoS Computational Biology* 9(8),

e1003118.

[19] Liang K-Y, Zeger SL (1986) Longitudinal data analysis using generalized linear models. *Biometrika* 73(1), 13–22.

[20] Ludington AJ, Hammond JM, Breen J, Deveson IW, Sanders KL (2023) New chromosome-scale genomes provide insights into marine adaptations of sea snakes (*Hydrophis*: Elapidae). *BMC Biology* 21(1), 284.

[21] Ma J, Devos KM, Bennetzen JL (2004) Analyses of LTR-retrotransposon structures reveal recent and rapid genomic DNA loss in rice. *Genome Research* 14(5), 860–869.

[22] Matsubara K, Sarre SD, Georges A, Matsuda Y, Marshall Graves JA, Ezaz T (2014) Highly differentiated ZW sex microchromosomes in the Australian *Varanus* species evolved through rapid amplification of repetitive sequences. *PLoS ONE* 9(4), e95226.

[23] Mu W, Davis ES, Lee S, Dozmorov MG, Phanstiel DH, Love MI (2023) bootRanges: flexible generation of null sets of genomic ranges for hypothesis testing. *Bioinformatics* 39(5), btad190.

[24] Niimura Y, Nei M (2006) Evolutionary dynamics of olfactory and other chemosensory receptor genes in vertebrates. *Journal of Human Genetics* 51(6), 505–517.

[25] O'Hara RB, Kotze DJ (2010) Do not log-transform count data. *Methods in Ecology and Evolution* 1(2), 118–122.

[26] Olvera-Vazquez SG, Chen X, Mesnil A, Meslin C, Almeida-Silva F, Confais J, Bourgeois Y, Lombardi G, Lougmani C, Alix K, Francillonne N, Choisne N, Cauet S, Simon J-C, Buchard C, Rodde N, Ogereau D, Mottet C, Degrave A, Segura E, Carbone A, Barrès B, Jacquin-Joly E, Marande W, Lavenier D, Legeai F, Cornille A (2025) Comprehensive annotation of olfactory and gustatory receptor genes and transposable elements revealed their evolutionary dynamics in aphids. *Molecular Biology and Evolution* 42(12), msaf238.

[27] Ou S, Chen J, Jiang N (2018) Assessing genome assembly quality using the LTR Assembly Index (LAI). *Nucleic Acids Research* 46(21), e126.

[28] Ou S, Su W, Liao Y, Chougule K, Agda JRA, Hellinga AJ, Lugo CSB, Elliott TA, Ware D, Peterson T, Jiang N, Hirsch CN, Hufford MB (2019) Benchmarking transposable element annotation methods for creation of a streamlined, comprehensive pipeline. *Genome Biology* 20(1), 275.

[29] Pasquesi GIM, Adams RH, Card DC, Schield DR, Corbin AB, Perry BW, Reyes-Velasco J, Ruggiero RP, Vandewege MW, Shortt JA, Castoe TA (2018) Squamate reptiles challenge paradigms of genomic repeat element evolution set by birds and mammals. *Nature Communications* 9(1), 2774.

[30] Pockrandt C, Alzamel M, Iliopoulos CS, Reinert K (2020) GenMap: ultra-fast computation of genome mappability. *Bioinformatics* 36(12), 3687–3692.

[31] Qian J, Xue H, Ou S, Mann L, Storer J, Fürtauer L, Heitkam T, Wildermuth MC, Kusch S,

Panstruga R (2025) TEtrimmer: a tool to automate the manual curation of transposable elements. *Nature Communications* 16(1), 8429.

[32] Quinlan AR, Hall IM (2010) BEDTools: a flexible suite of utilities for comparing genomic features. *Bioinformatics* 26(6), 841–842.

[33] Rao W-Q, Rivera-de-Torre E, Seneci L, Shi M-H, Zhang Y-L, Lin L, Lan T-M, Pungerčar J, Liu S-Q, Laustsen AH (2025) A chromosome-level genome assembly of the western nose-horned viper (*Vipera ammodytes ammodytes*). *Genome Biology and Evolution* 17(12), evaf210.

[34] Rapoport R, Greenberg A, Yakhini Z, Simon I (2024) A cyclic permutation approach to removing spatial dependency between clustered gene ontology terms. *Biology* 13(3), 175.

[35] Sanders KL, Lee MSY, Mumpuni, Bertozzi T, Rasmussen AR (2013) "Multilocus phylogeny and recent rapid radiation of the viviparous sea snakes (Elapidae: Hydrophiinae)" *Molecular Phylogenetics and Evolution* 66, 575–591. doi:10.1016/j.ympev.2012.09.021

[36] Schield DR, Perry BW, Card DC, Pasquesi GIM, Westfall AK, Mackessy SP, Castoe TA (2022) The rattlesnake W chromosome: a GC-rich retroelement refugium with retained gene function across ancient evolutionary strata. *Genome Biology and Evolution* 14(9), evac116.

[37] Tarailo-Graovac M, Chen N (2009) Using RepeatMasker to identify repetitive elements in genomic sequences. *Current Protocols in Bioinformatics* 25(1), 4.10.1–4.10.14.

[38] Traherne JA, Martin M, Ward R, Ohashi M, Pellett F, Gladman D, Middleton D, Carrington M, Trowsdale J (2010) Mechanisms of copy number variation and hybrid gene formation in the KIR immune gene complex. *Human Molecular Genetics* 19(5), 737–751.

[39] Waples RS, Waples RK, Ward EJ (2022) Pseudoreplication in genomic-scale data sets. *Molecular Ecology Resources* 22(2), 503–518.

[40] Zhang R-G, Li G-Y, Wang X-L, Dainat J, Wang Z-X, Ou S, Ma Y (2022) TEsorter: an accurate and fast method to classify LTR-retrotransposons in plant genomes. *Horticulture Research* 9, uhac017.

[41] Zhu Z, Dobry J, Wapstra E, Zhou Q, Ezaz T (2026) Gene traffic mediated by transposable elements shaped the dynamic evolution of ancient sex chromosomes of varanid lizard. *Journal of Genetics and Genomics* 53(3), 497–509.